

Welcome to CS 3250 Software Testing

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Have You Ever Wondered



[Ref: emoji by Ekarin Apirakthanakorn]

Why This Course

Testing software properly is a fundamental skill in modern software development

Software testing is a crucial activity to verify that programs fulfill requirements and to ensure software compliance

Software testing is about engineering tests that detect problems in software

Testing is so much more than writing a bunch of JUnit or Selenium tests

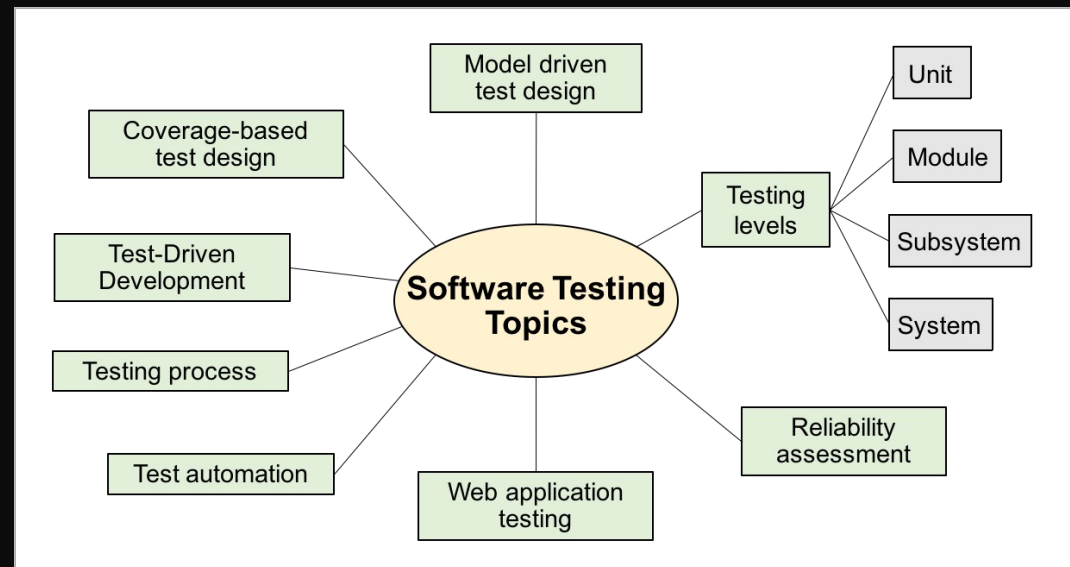
*Software testing is about **designing good and effective tests** under time and budget constraints*



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What This Course is About

- Importance of Software Testing
- Testing in Practice: design, automation, execution, evaluation
- Test-Driven Development
- Test Coverage Criteria
 - Input space partitioning
 - Graph coverage
 - Logic coverage
 - Syntax coverage
- Web app testing
- Bypass testing



“The true subject matter of the tester is not testing, but the design of test cases” —Jeff Offutt

Instead of how testing **is done**,
we focus on
how it **should be done** and
how it **will be done**

Our Focus

Designing effective tests

- How test engineers design tests
 - Explore a variety of test design concepts
 - Apply the theories to design tests
 - Automate tests using JUnit and Selenium
 - Be exposed to challenges and explore best practices in test automation
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Principles over syntax

- Foundational principles rather than specific programming syntax
 - Does not train specific test automation or software development frameworks
 - Provide the fundamentals needed to explore any automation framework independently
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Role of generative AI

- Does not teach how to use AI
- Focus on not blindly relying on AI outputs
- Explore, analyze, assess, and revise AI outputs to facilitate the testing process

You Will be Able to Answer

- What is software testing?
- Why do we test software?
- When should we test software?
- Who should test software?
- How should we test software and why should we do it that way?
- When should we stop testing? Good enough?
- How many tests do we need to write?
- How do we choose test inputs effectively?
- How can we measure the quality of a test suite?
- How can we improve the quality of a test suite?
- How can we reduce the testing cost?



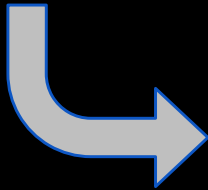
I Hope to Help You

Test software adequately, appropriately, efficiently, and effectively

Be able to lead and work effectively in collaborative environments

Have confidence in learning and applying new testing approaches or frameworks to test any software

Get excited about exploring and developing new testing approaches to appropriately test any software



Become better testers

- Create high-quality tests at all testing levels
- Understand practical ways to design and automate tests
- Apply theory in practical ways

Become better programmers

- Be aware of potential problems in software and create high-quality developer tests
- Develop maintainable and testable software

Become better engineers

- Build programs and test them in a unified manner

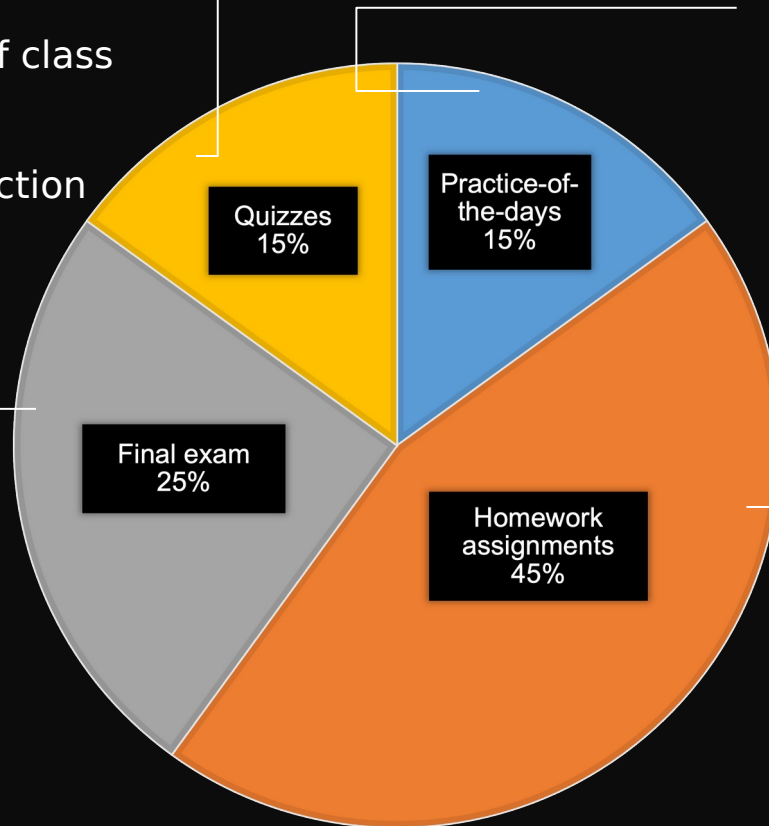
Become better thinkers

- Solve problems in logical and analytical ways

What You will Do & How Your Learning will be Evaluated

Quizzes:

- 5 quizzes
- First 15 minutes of class
- Close book/note
- In class
- Reflection & correction



Practice-of-the-days:

- Weekly
- Grade on done/not done
- Due 1pm EST the next day
- -25% for 24 hours late; -50% for 48 hours late
- Not accepted after 48 hours past due

Final exam:

- Comprehensive
- Close book/note
- 1 handwritten note sheet
- In class

Homework assignments:

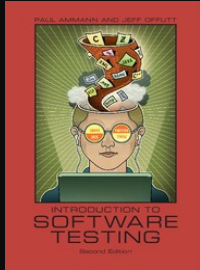
- Almost weekly
- Due before class
- Early-submission bonus points
- -25% for 24 hours late; -50% for 48 hours late
- Not accepted after 48 hours past due

Hands-on activities:

- Almost every meeting
- No submission

More Info to Help You Learn

- Class URL: <http://www.cs.virginia.edu/~up3f/cs3250/>
- Readings: Check schedule page, please read before class
- Textbook:



(book website)

<https://www.albany.edu/faculty/offutt/softwaretest/>

(student solution)

<https://www.albany.edu/faculty/offutt/softwaretest/exer-student.pdf>

- Emails: Use UVA email, check it regularly, include “CS 3250”

Share Ideas, Get Help, Just Talk

- Instructor office hours: days/times on class website
- Piazza: <https://piazza.com/virginia/fall2026/cs3250>
- Questions should be posted to an appropriate thread
 - Answered by instructor and your peers
 - Public: general questions and answers
 - Private: Grade/homework-specific questions – instructor and TAs
- Do not email general homework or class questions, **use Piazza**

Learn with *“Purpose”* and *“Vision”*

Help yourself learn: <https://forms.office.com/r/uwbbLJEGQT>

- To better support and emphasize your differences, energy, and passions, as well as to enhance your strengths
- You are encouraged to be involved in designing and shaping this course to maximize your learning experience, skills, and knowledge
- Please feel free to share your ideas / opinions / suggestions on things that we should do or try to help you learn software testing, using the “Help yourself learn” form

Additional Things to Know

- Submission site: Gradescope
<https://www.gradescope.com/courses/1341917>
- Regrade request: Submit to Gradescope
 - Regrade the “entire” work
 - Correct any grading mistakes
 - May result in either an increase or a decrease in grade
 - Not try to scare off students whose work was graded incorrectly, but try to avoid frivolous requests
- Check schedule page regularly
 - In case of emergency, email & post update on the schedule page

Prerequisites

- CS 2120 and (CS 2100 or place out test) with C- or better
- Discrete math, data representation, (some) software engineering
- Java, syntax and semantics of multiple programming languages
- Programming skills
- Software installation and troubleshooting skills
- **Willingness** and **excitement** to learn new concepts and experience new frameworks

Your  and soul !!

*Prerequisites define what you need to know **before** taking a class to succeed in the class. Please do not ask if you can take the class without the prerequisites, I have to advise **against** it*

The CS student services staff will monitor and remove students not meeting prerequisites in courses in which they are enrolled

Accommodations

<https://www.cs.virginia.edu/~up3f/cs3250/policies.html#no-extension>

Accommodations: Religious

If you wish to request academic accommodation for religious observance, please let me know via email as far in advance as possible.

If there exists a religious conflict, please mention it.

Do not say *"You are busy"* or *"You have a busy start to the week."* *"Being busy / having a busy schedule / having a busy start to the week"* is **not** a sufficient justification for an accommodation / extension.

"Being busy / having a busy schedule / having a busy start to the week" does not imply a religious conflict.

If you need religious accommodations, please say that you need religious accommodations.

Accommodations: Schedule conflict

If there exists a schedule conflict, please provide the necessary information to determine an accommodation.

Do not say *"You are busy"* or *"You have an unavoidable conflict."* *"Being busy / having a busy schedule / having an unavoidable conflict"* is **not** a sufficient justification for an accommodation / extension.

It is important to note that being busy with other courses (homework assignments, projects, exams, quizzes, tests, ...) is **not** a sufficient justification for an accommodation / extension. Please consider adjusting or planning your time management.

Accommodations: Things in life and special circumstances

If you believe that circumstances warrant an extension or exception, come talk to the course instructor and we will resolve the situation as best we can.